

EE619: Assignment 1 – SPICE Simulations

Name Mahamakli Murari

Roll No. 230616

(Write your roll no. on space provided on all sheets. Penalty of -2 marks if the roll no. is missing.)

A1.

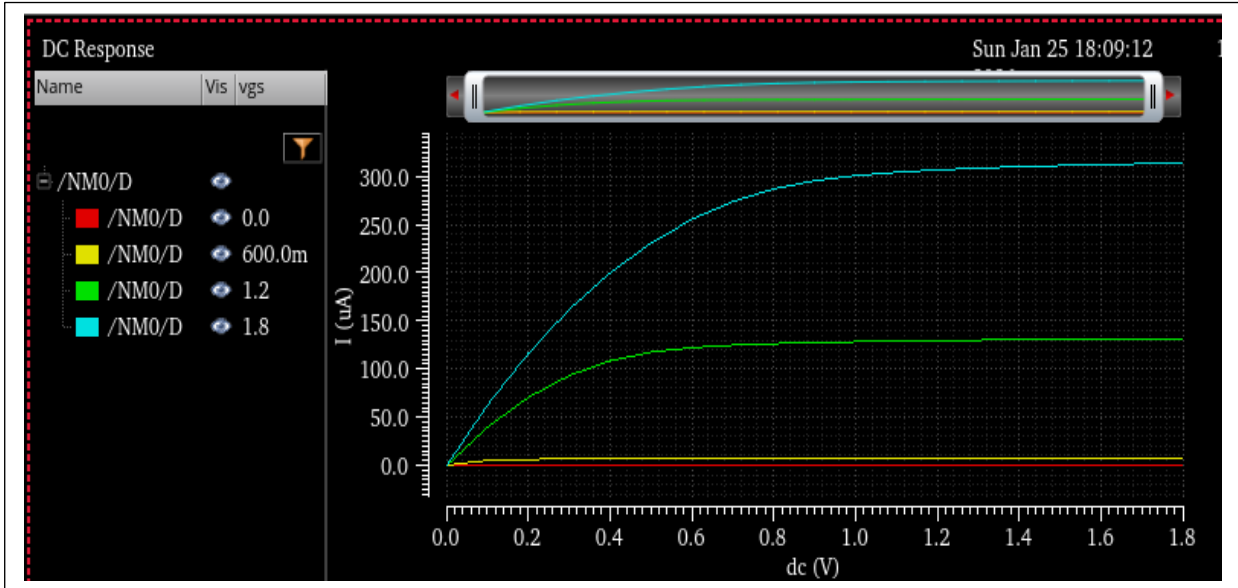


Fig. 1 (X axis- $V_{ds}(V)$, Y axis – $I_{ds}(uA)$)

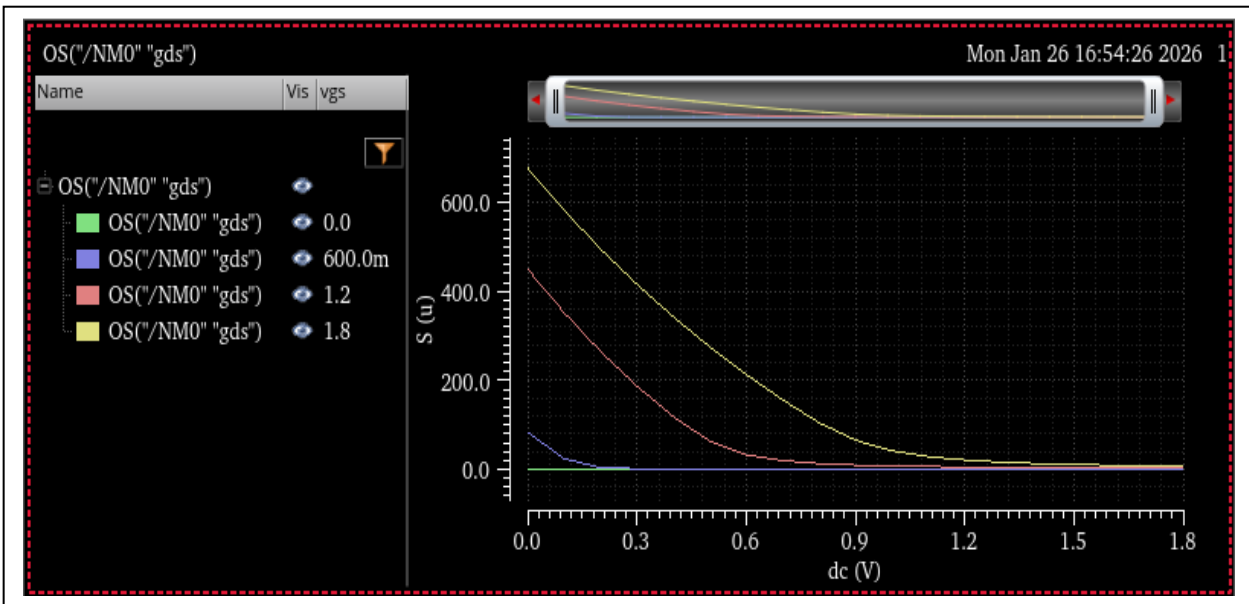


Fig. 2 (X axis- $V_{ds}(V)$, Y axis – $g_{ds}(uS)$)

Observation:

The output conductance g_{ds} is high at low values of V_{DS} , indicating that the NMOS transistor operates in the linear (ohmic) region. As V_{DS} increases, the device enters saturation and the value of g_{ds} decreases significantly. However, g_{ds} does not become zero in saturation due to channel length modulation. Higher values of V_{GS} result in slightly higher output conductance in saturation because of increased drain current. This finite output conductance limits the output resistance of the NMOS transistor.

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A2.

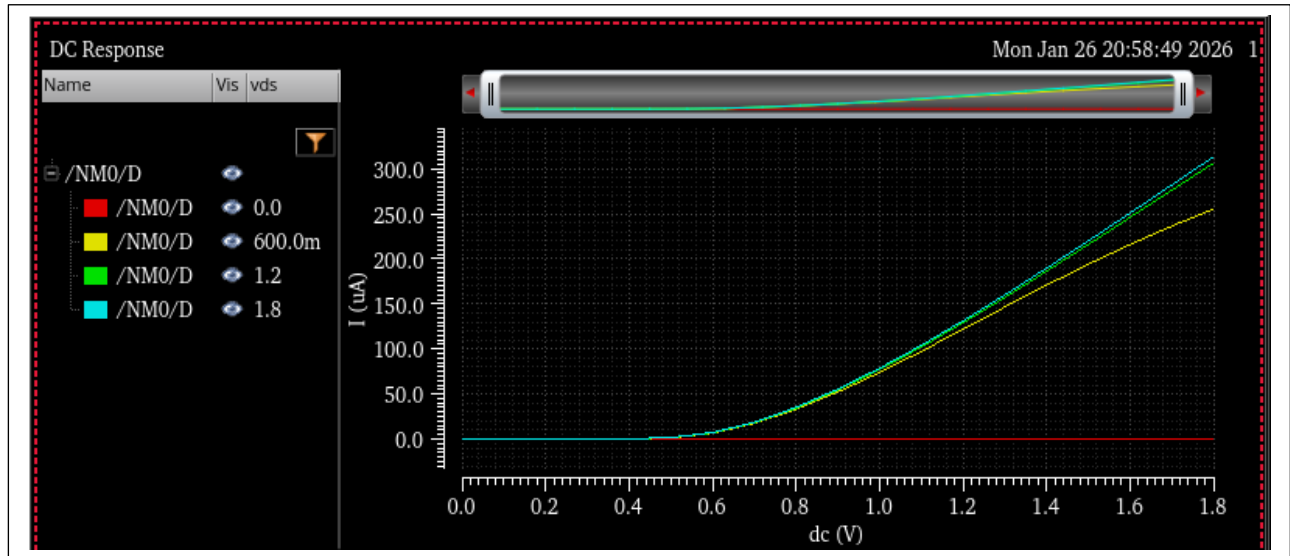


Fig. 3 (X axis- $V_{gs}(V)$, Y axis – $I_{ds}(uA)$)

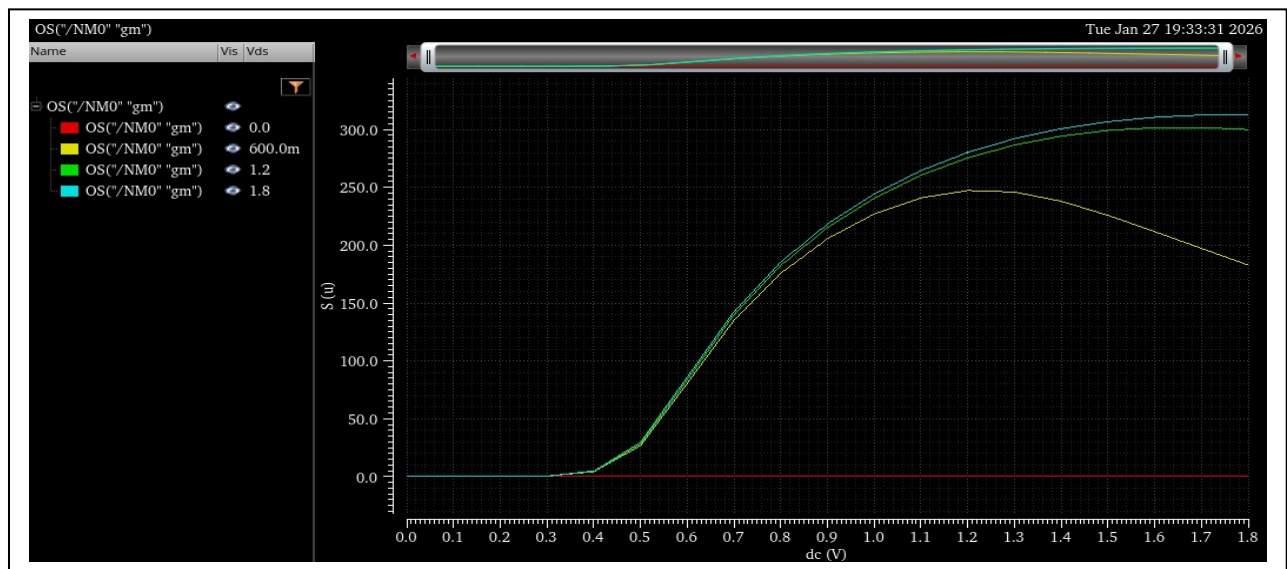


Fig. 4 (X axis- $V_{gs}(V)$, Y axis – $g_m(us)$)

Observation:

The transconductance g_m remains nearly zero when the gate-to-source voltage V_{GS} is below the threshold voltage V_T , as the transistor is in cutoff. Once V_{GS} exceeds V_T , g_m increases rapidly, indicating strong inversion and efficient channel formation. With an increase in V_{DS} , the peak value of transconductance also increases. At higher values of V_{GS} , the rate of increase of g_m reduces due to mobility degradation and other non-ideal effects. Transconductance is a key parameter that determines the gain capability of the NMOS transistor.

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A3.

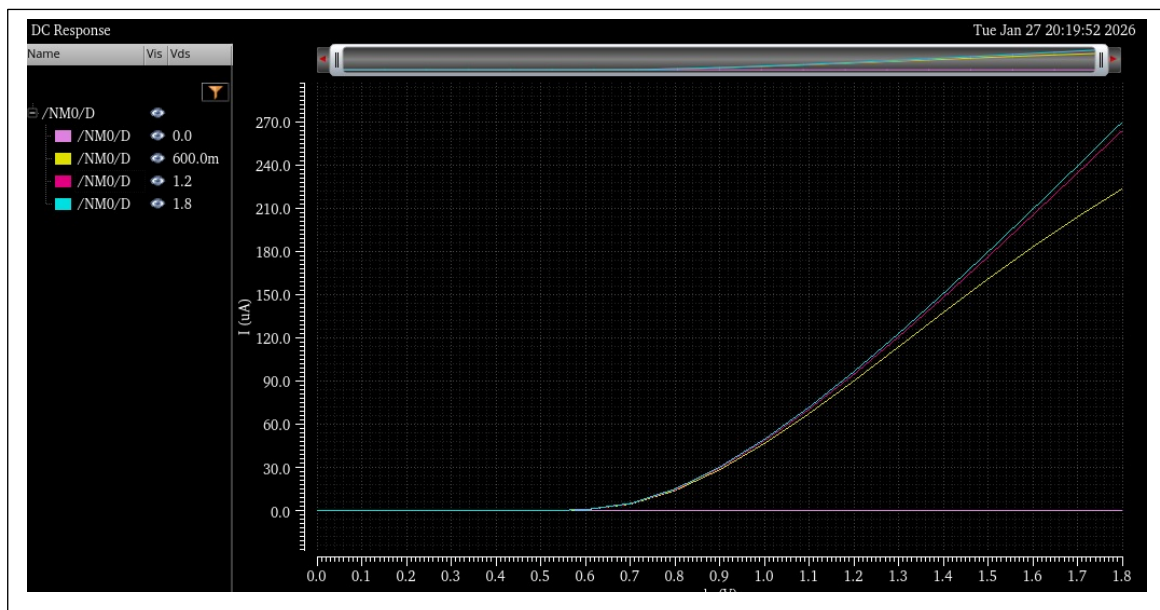


Fig. 5 (X axis- $V_{gs}(V)$, Y axis – $I_{ds}(\mu A)$) 0.5 V_{sb}

Observation:

When a substrate bias of $V_{SB} = 0.5 V$ is applied, the threshold voltage of the NMOS transistor increases due to the body effect. As a result, the transfer characteristics shift towards higher values of V_{GS} compared to the case where $V_{SB} = 0 V$. For the same gate-to-source voltage, the drain current is lower when substrate bias is present. This reduction in drain current occurs because reverse body bias reduces the channel charge density. The body effect therefore degrades the current drive capability and must be considered in practical circuit design.

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A4.

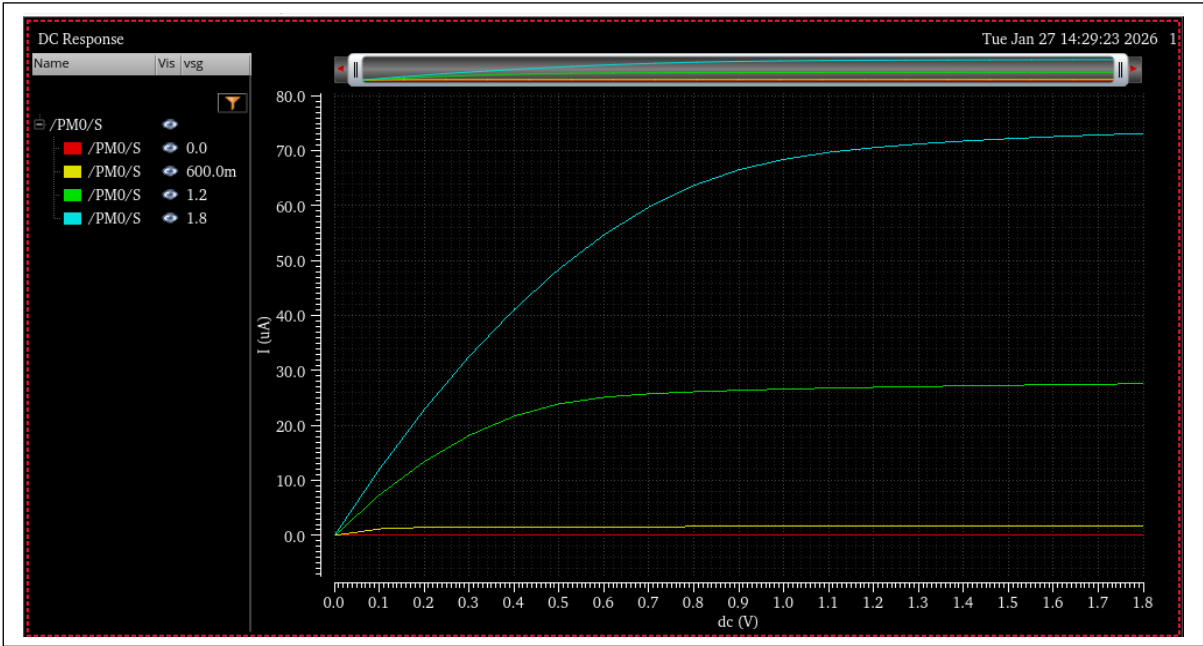


Fig. 6 Pmos ((X axis- $V_{sd}(V)$, Y axis – $I_{sd}(\mu A)$))

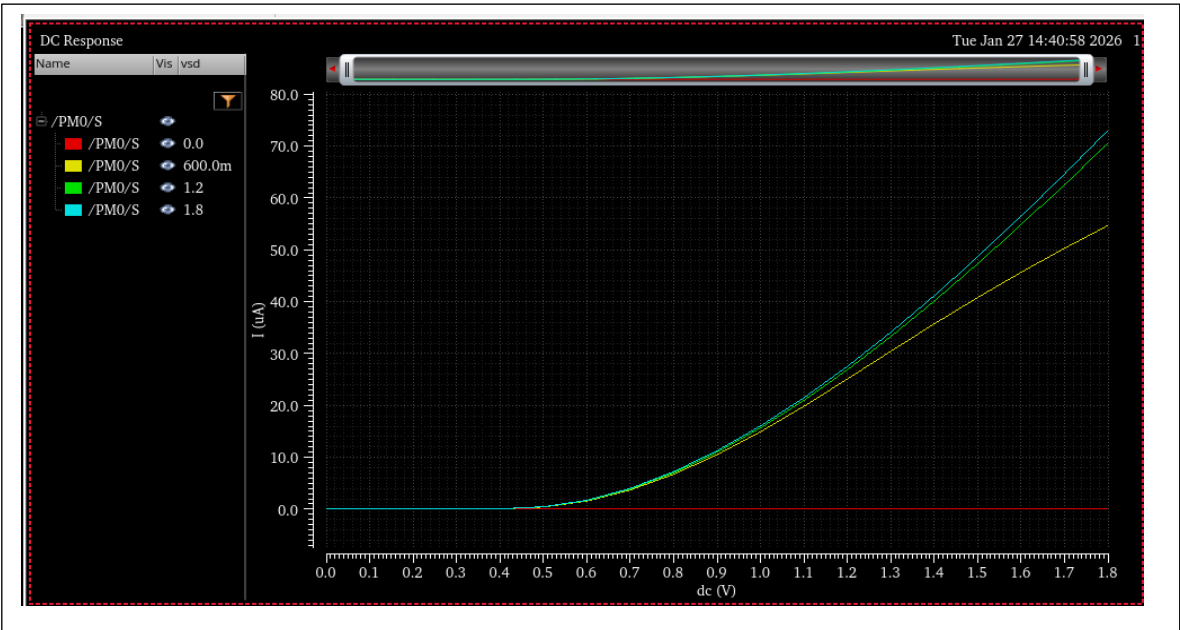


Fig. 7 Pmos ((X axis- $V_{sg}(V)$, Y axis – $I_{sd}(\mu A)$))

Observation:

Output and transfer characteristics of PMOS are similar to that of NMOS.

